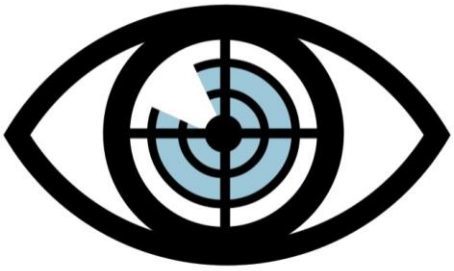
The University of Texas Rio Grande Valley

CYBI: Networking & Data Communications Lab 5

Network Mapping with NMAP



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I. Introduction

In this lab, you will use the Nmap (Nmap, short for Network Mapper), a network discovery and security auditing tool to scan for:

- Open ports and services (transport layer ports and their associated service)
- Discover services along with their versions
- Guess the operating system running on a target machine
- Get accurate packet routes till the target machine
- Monitoring hosts

II. NMAP Port Types

The six port states recognized and reported by Nmap are:

open

An application is actively accepting TCP connections, UDP datagrams or SCTP associations on this port. Finding these is often the primary goal of port scanning. Security-minded people know that each open port is an avenue for attack. Attackers and pen-testers want to exploit the open ports, while administrators try to close or protect them with firewalls without thwarting legitimate users. Open ports are also interesting for non-security scans because they show services available for use on the network.

closed

A closed port is accessible (it receives and responds to Nmap probe packets), but there is no application listening on it. They can be helpful in showing that a host is up on an IP address (host discovery, or ping scanning), and as part of OS detection. Because closed ports are reachable, it may be worth scanning later in case some open up. Administrators may want to consider blocking such ports with a firewall. Then they would appear in the filtered state, discussed next.

filtered

Nmap cannot determine whether the port is open because ***packet filtering prevents*** its probes from reaching the port. The filtering could be from a dedicated firewall device, router rules, or host-based firewall software. These ports frustrate attackers because they provide so little information. Sometimes, they respond with ICMP error messages such as type 3 code 13 (destination unreachable: communication administratively prohibited), but filters that simply drop probes without responding are far more common. This forces Nmap to retry several times just in case the probe was dropped due to network congestion rather than filtering. This slows down the scan dramatically.

unfiltered

The unfiltered state means that ***a port is accessible***, but Nmap is unable to determine whether it is open or closed. Only the ACK scan, which is used to map firewall rulesets, classifies ports into this state. Scanning unfiltered ports with other scan types such as Window scan, SYN scan, or FIN scan, may help resolve whether the port is open.

open|filtered

Nmap places ports in this state when it is unable to determine whether a port is open or filtered. This occurs for scan types in which open ports give no response. The lack of response could also mean that a packet filter dropped the probe or any response it elicited. So Nmap does not know for sure whether the port is open or being filtered. The UDP, IP protocol, FIN, NULL, and Xmas scans classify ports this way.

closed|filtered

This state is used when Nmap is unable to determine whether a port is closed or filtered. It is only used for the IP ID idle scan.

III. Installation

Nmap and Zenmap (the graphical front end) are available in several versions and formats. Use https://nmap.org/download.html to download the windows version.

Microsoft Windows binaries

Please read the Windows section of the Install Guide for limitations and installation instructions for the Windows version of Nmap. You can choose from a self-installer (includes dependencies and also the Zenmap GUI) or the much smaller command-line zip file version. We support Nmap on Windows 7 and newer, as well as Windows Server 2008 and newer. We also maintain a guide for users who must run Nmap on earlier Windows releases..

The Nmap **executable Windows installer** can handle Npcap installation, registry performance tweaks, and decompressing the executables and data files into your preferred location. It also includes the Zenmap graphical frontend. Skip all the complexity of the Windows zip files with a self-installer:

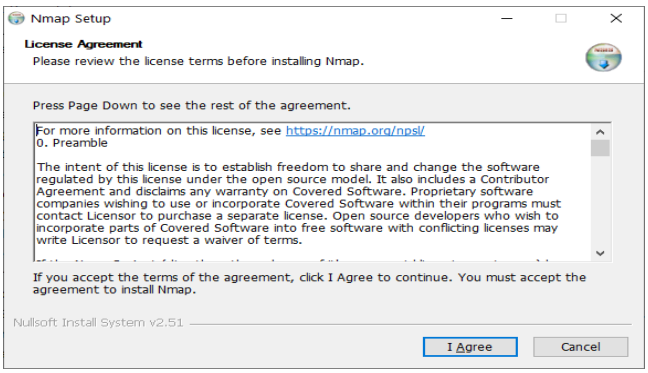
Latest stable release self-installer: nmap-7.91-setup.exe

We have written post-install usage instructions. Please notify us if you encounter any problems or have suggestions for the installer.

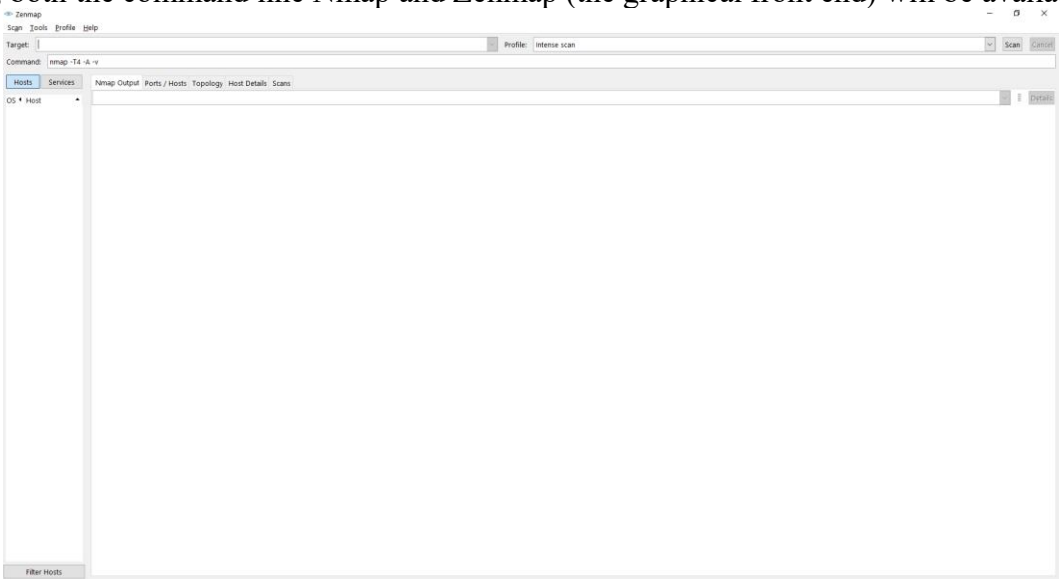
For those who prefer the command-line zip files (Installation Instructions; Usage Instructions), they are still available. The Zenmap graphical interface is *not* included with these, so you need to run nmap.exe from a DOS/command window. Or you can download and install a superior command shell such as those included with the free Cygwin system. Also, you need to run the Npcap and Microsoft Visual C++ 2013 Redistributable Package installers which are included in the zip file. The main advantage is that these zip files are a fraction of the size of the executable installer:

Latest stable command-line zipfile: nmap-7.91-win32.zip

Run thenmap-7.91-setup.exe (accept all defaults)



Once installed, both the command line Nmap and Zenmap (the graphical front end) will be available.



Zenmap GUI

```
PS C:\Users\christopher> nmap
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN, TCP ACK, UDP or SCTP discovery to given por
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
```

Command line Nmap

In this lab, we will focus on the command line as it allows for more complex options and customization

IV. Basic Scan Types -sT and -sS

The two basic scan types used most in Nmap are TCP connect() scanning [-sT] and SYN scanning (also known as half-open, or ***stealth scanning***) [-sS]. Once the scan is completed, ports to which a connection could be established are listed as open, the rest are said to be closed. Open means that you can connect to the port. Firewall or intrusion detection system can trigger a warning and servers will log connections and their source IP, so it would be easy to detect the source of a TCP connect() scan. However, with ***SYN or Stealth scanning*** this is avoided because it makes use of the 3-way handshake procedure by sending a SYN packet and looking at the response. If SYN/ACK is sent back, the port is open, and the remote end is trying to open a TCP connection. The scanner then sends an RST to tear down the connection before it can be established fully; often preventing the connection attempt of appearing in any application logs.

1. The results for the -sS scan for scanme.nmap.org isshown below. Note the output format (***Port State Service***). Many of these ports/services should be familiar to you by now.
2. The results for the -sS scan for www.utrgv.edu and gaia.cs.umass.edu are shown below. We can see that they netted more ports/services than the T scan.

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\christopher> nmap -sT scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:04 Central Daylight Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.057s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  ssh
80/tcp    open  http
9929/tcp  open  nping-echo
31337/tcp open  Elite

Nmap done: 1 IP address (1 host up) scanned in 12.22 seconds
PS C:\Users\christopher>
```

V. FIN, Null and Xmas Tree Scans [-sF, -sN, -sX]

Each scan type refers to the ***flags set in the TCP header***. The idea behind these types of scans is that a closed port should respond with an RST upon receiving packets, whereas an open port should just drop them (it's listening for packets with SYN set). The ***FIN*** scan sends a packet with only the FIN flag set, the ***Xmas*** Tree scan sets the FIN, URG and PUSH flags, and the ***Null*** scan sends a packet with no flags switched on. Most of the time, the results of the -sS and -sF are the same, but it is less likely for the -sF to show up in a logging system.

```
Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\christopher> nmap -sF scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:12 Central Daylight Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.055s latency).
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 12.48 seconds
PS C:\Users\christopher>
```

```
PS C:\Users\christopher> nmap -sX scanmen.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:13 Central Daylight Time
Nmap scan report for scanmen.nmap.org (50.116.1.184)
Host is up (0.055s latency).
rDNS record for 50.116.1.184: ack.nmap.org
All 1000 scanned ports on scanmen.nmap.org (50.116.1.184) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 12.58 seconds
PS C:\Users\christopher>
```

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\christopher> nmap -sN scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:14 Central Daylight Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.055s latency).
All 1000 scanned ports on scanme.nmap.org (45.33.32.156) are in ignored states.
Not shown: 1000 open|filtered tcp ports (no-response)

Nmap done: 1 IP address (1 host up) scanned in 12.52 seconds
PS C:\Users\christopher>
```

VI. Ping Scan [-sP]

This scan type lists ***the hosts*** within the specified range that responded to a ping. It allows you to detect ***which computers are online, rather than which ports are open***. Four methods exist within Nmap for ***ping sweeping***. The first method sends an ICMP ECHO REQUEST (ping request) packet to the destination system. If an ICMP ECHO REPLY is received, the system is up, and ICMP packets are not blocked. If there is no response to the ICMP ping, Nmap will try a "TCP Ping", to determine whether ICMP is blocked, or if the host is really not online. The ICMP scan type can be disabled by setting -P0 (that is, zero, not uppercase o) option. Note that you can supply a network instead of one single host in all the Nmap commands. For example, any of these should work

scanme.nmap.org
microsoft.com/24
192.168.0.1
10.0.0-255.1-254

1. The results for the -sP on my network at home. I ran it with the range 192.168.1.1-254. Run the ipconfig /all to get your network addresses. I am not showing the results as it was showing the name of each device and the MAC addresses, and other information.

```
PS C:\Users\christopher> nmap -sP 192.168.1.1-254
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:27 Central Daylight Time
Stats: 0:02:11 elapsed; 0 hosts completed (0 up), 254 undergoing Ping Scan
Ping Scan Timing: About 63.98% done; ETC: 18:31 (0:01:14 remaining)
Stats: 0:02:11 elapsed; 0 hosts completed (0 up), 254 undergoing Ping Scan
Ping Scan Timing: About 64.42% done; ETC: 18:31 (0:01:12 remaining)
```

The results for the -sP on scanme.nmap.org

```
PS C:\Users\christopher> nmap -sP scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:32 Central Daylight Time
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.056s latency).
Nmap done: 1 IP address (1 host up) scanned in 5.92 seconds
PS C:\Users\christopher> nmap -sP scanme.nmap.org/24
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:33 Central Daylight Time
Nmap scan report for 45.33.32.1
Host is up (0.060s latency).
Nmap scan report for 45.33.32.3
Host is up (0.060s latency).
Nmap scan report for business-software.shop (45.33.32.4)
Host is up (0.053s latency).
Nmap scan report for 45-33-32-5.ip.linodeusercontent.com (45.33.32.5)
Host is up (0.061s latency).
Nmap scan report for 45-33-32-8.ip.linodeusercontent.com (45.33.32.8)
Host is up (0.055s latency).
Nmap scan report for li982-9.members.linode.com (45.33.32.9)
Host is up (0.055s latency).
Nmap scan report for fivekeys.issoasis.com (45.33.32.10)
Host is up (0.055s latency).
```

VII. UDP Scan [-sU]

Scanning for open UDP ports is done with the -sU option. With this scan type, Nmap sends 0-byte UDP packets to each target port on the victim. Receipt of an ICMP Port Unreachable message signifies the port is closed, otherwise it is assumed open. One major problem with this technique is that, when a firewall blocks outgoing ICMP Port Unreachable messages, the port will appear open. These false positives are hard to distinguish from real open ports. Most modern services utilise TCP, and thus UDP scanning is not usually included in a pre-attack information gathering exercise unless a TCP scan or other sources indicate that it would be worth the time taken to perform a UDP scan.

1. The results for the -sP on google DNS servers.

```
PS C:\Users\christopher> nmap -sU 8.8.4.4
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:36 Central Daylight Time
Nmap scan report for dns.google (8.8.4.4)
Host is up (0.016s latency).
Not shown: 998 open|filtered udp ports (no-response)
PORT      STATE  SERVICE
53/udp    open   domain
33459/udp closed unknown

Nmap done: 1 IP address (1 host up) scanned in 10.45 seconds
PS C:\Users\christopher> nmap -sU 8.8.8.8
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 18:37 Central Daylight Time
Nmap scan report for dns.google (8.8.8.8)
Host is up (0.017s latency).
Not shown: 998 open|filtered udp ports (no-response)
PORT      STATE  SERVICE
53/udp    open   domain
33459/udp closed unknown

Nmap done: 1 IP address (1 host up) scanned in 10.38 seconds
PS C:\Users\christopher>
```

VIII. IP Protocol Scans [-sO]

The IP Protocol Scans attempt to determine the IP protocols supported on a target. Nmap sends a raw IP packet without any additional protocol header, to each protocol on the target machine. Receipt of an ***ICMP Protocol Unreachable*** message tells us the protocol is not in use, otherwise it is assumed open. Not all hosts send ICMP Protocol Unreachable messages. These may include firewalls, AIX, HP-UX and Digital UNIX. These machines will report all protocols open.

1. The results for the -sO on my Computer

```
PS C:\Users\christopher> nmap -sO 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:00 Central Daylight Time
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0015s latency).
Not shown: 249 closed n/a protocols (proto-unreach)
PROTOCOL STATE         SERVICE
1        open          icmp
2        open|filtered igmp
6        open          tcp
17       open          udp
50       open|filtered esp
51       open|filtered ah
58       open|filtered ipv6-icmp
```

IX. IP Protocol Scans [-sO]

The IP Protocol Scans attempt to determine the IP protocols supported on a target. Nmap sends a raw IP packet without any additional protocol header, to each protocol on the target machine. Receipt of an ***ICMP Protocol Unreachable*** message tells us the protocol is not in use, otherwise it is assumed open. Not all hosts send ICMP Protocol Unreachable messages. These may include firewalls, AIX, HP-UX and Digital UNIX. These machines will report all protocols open.

1. The results for the -sO on my Computer

```
PS C:\Users\christopher> nmap -sO 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:00 Central Daylight Time
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0015s latency).
Not shown: 249 closed n/a protocols (proto-unreach)
PROTOCOL STATE         SERVICE
1        open          icmp
2        open|filtered igmp
6        open          tcp
17       open          udp
50       open|filtered esp
51       open|filtered ah
58       open|filtered ipv6-icmp
```

X. Timing and Hiding Scans

Nmap adjusts its timings automatically depending on network speed and response times of the victim. The main timing option is set through the -T parameter. There are six predefined timing policies which can be specified by name or number (starting with 0, corresponding to Paranoid timing). The timings are **Paranoid**, **Sneaky**, **Polite**, **Normal**, **Aggressive** and **Insane**.

A -T Paranoid (or -T0) scan will wait (generally***) at least 5 minutes between each packet sent*** to fool the firewall.

A -T Insane (or -T5) scan will map a host in very little time.

Timings for individual aspects of a scan can also be set using the --host timeout, --max rtt timeout, --min rtt timeout, --initial rtt timeout, --max parallelism, --min parallelism, and --scan delay options. Used for example to give up on slow target hosts.

```
PS C:\Users\christopher> nmap -T5 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:03 Central Daylight Time
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.00015s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.90 seconds
PS C:\Users\christopher> nmap -T4 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:03 Central Daylight Time
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.90 seconds
```

XI. Other Scans

- **Idle** Scanning [-sI]: Idle scanning is an advanced, highly stealthed technique, where no packets aresent to the target which can be identified to originate from the scanning machine
- **Version** Detection [-sV]; ersion Detection collects information about the specific service running on an open port, including the product name and version number
- **ACK** Scan [-sA]: Usually used to map firewall rulesets and distinguish between stateful and stateless firewalls, this scan type sends ACK packets to a host

XII. OS Detection

The inner workings of the operating System (OS) detection are quite complex, but it is one of the easiest features to use. Simply add -O to your scan options. You may want to also increase the verbosity with -v for even more OS-related details

1. The results for running against the scanme.nmap.org Computer. I ran it against my computer and it gave all the details as well!!

```
PS C:\Users\christopher> nmap -O -v 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:13 Central Daylight Time
Initiating Parallel DNS resolution of 1 host. at 19:13
Completed Parallel DNS resolution of 1 host. at 19:13, 5.51s elapsed
Initiating SYN Stealth Scan at 19:13
Scanning WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227) [1000 ports]
Discovered open port 135/tcp on 129.113.104.227
Discovered open port 445/tcp on 129.113.104.227
Discovered open port 139/tcp on 129.113.104.227
Completed SYN Stealth Scan at 19:13, 0.10s elapsed (1000 total ports)
Initiating OS detection (try #1) against WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.00038s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
Device type: general purpose
Running: Microsoft Windows 10|11
OS CPE: cpe:/o:microsoft:windows_10 cpe:/o:microsoft:windows_11
OS details: Microsoft Windows 10 1607 - 11 23H2
Uptime guess: 11.159 days (since Tue Jun 18 15:25:16 2024)
Network Distance: 0 hops
TCP Sequence Prediction: Difficulty=257 (Good luck!)
IP ID Sequence Generation: Incremental
```

XIII. Outputting Logs

- Logging in Nmap can be provided by the -oN, -oX or -oG options. Each one is followed by the name of the logfile. -oN outputs a human readable log, -oX outputs an XML log and -oG outputs a grepable log. The -oA option outputs in all 3 formats
    - The –**append-output** option appends scan results to the output files you specified instead of overwriting their contents.

2. The results for the -oN on my Computer

```
PS C:\Users\christopher> nmap -T5 -oN nmap.log 129.113.104.227
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:16 Central Daylight Time
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Nmap done: 1 IP address (1 host up) scanned in 5.91 seconds
PS C:\Users\christopher> type nmap.log
# Nmap 7.95 scan initiated Sat Jun 29 19:16:02 2024 as: "C:\\Program Files (x86)\\Nmap\\nmap.exe" -T5 -oN nmap.log 129.113.104.227
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

# Nmap done at Sat Jun 29 19:16:08 2024 -- 1 IP address (1 host up) scanned in 5.91 seconds
```

3. The results for the -oN with the append option on my Computer

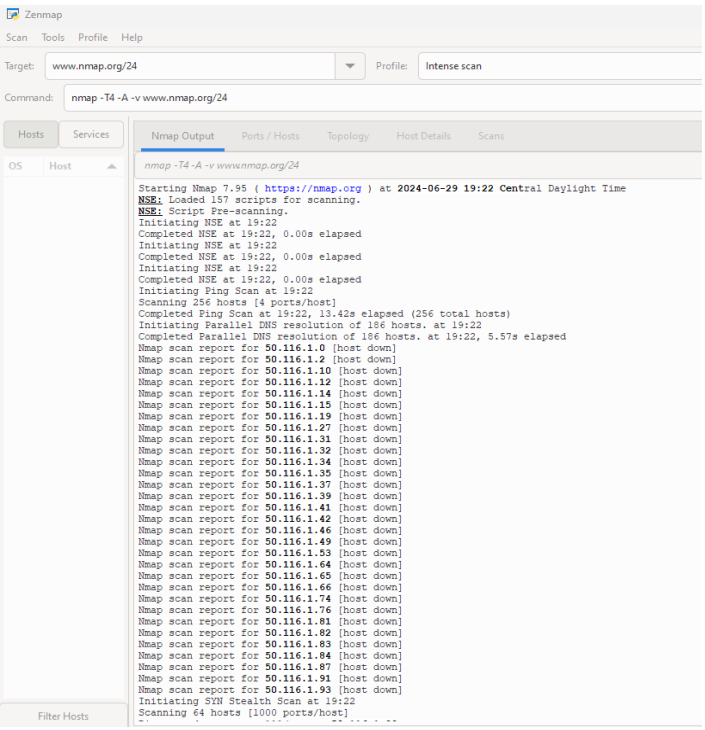
```
Nmap done: 1 IP address (1 host up) scanned in 5.92 seconds
PS C:\Users\christopher> nmap -T5 --append-output -oN nmap.log 129.113.104.227cls
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:18 Central Daylight Time
Failed to resolve "129.113.104.227cls".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.15 seconds
PS C:\Users\christopher>
```

```
PS C:\Users\christopher> nmap -T5 --append-output -oN nmap.log 129.113.104.227cls
Starting Nmap 7.95 ( https://nmap.org ) at 2024-06-29 19:19 Central Daylight Time
Failed to resolve "129.113.104.227cls".
WARNING: No targets were specified, so 0 hosts scanned.
Nmap done: 0 IP addresses (0 hosts up) scanned in 0.08 seconds
PS C:\Users\christopher> type nmap.log
# Nmap 7.95 scan initiated Sat Jun 29 19:16:02 2024 as: "C:\\Program Files (x86)\\Nmap\\nmap.exe" -T5 -oN nmap.log 129.113.104.227
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.0010s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

# Nmap done at Sat Jun 29 19:16:08 2024 -- 1 IP address (1 host up) scanned in 5.91 seconds
# Nmap 7.95 scan initiated Sat Jun 29 19:18:05 2024 as: "C:\\Program Files (x86)\\Nmap\\nmap.exe" -T5 --append-output -oN nmap.log 129.1
13.104.227
Nmap scan report for WIN-ER5P4AFAOUP.ad.utrgv.edu (129.113.104.227)
Host is up (0.00033s latency).
Not shown: 997 closed tcp ports (reset)
PORT    STATE SERVICE
135/tcp open  msrpc
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds
```

XIV. Running the GUI NMAP

1. The results running the gui with nmap.org as the targer. Check all the tabs as there are a lot of useful information. Here are two screen shots.



Appendix: NMAP Syntax

Cmd > nmap

Nmap 7.91 (https://nmap.org)
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
Can pass **hostnames, IP addresses, networks**, etc.

```
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
  --scanflags <flags>: Customize TCP scan flags
  -sI <zombie host[:probeport]>: Idle scan
  -sY/sZ: SCTP INIT/COOKIE-ECHO scans
  -sO: IP protocol scan
  -b <FTP relay host>: FTP bounce scan
PORT SPECIFICATION AND SCAN ORDER:
  -p <port ranges>: Only scan specified ports
    Ex: -p22; -p1-65535; -p U:53,111,137,T:21-25,80,139,8080,S:9
  --exclude-ports <port ranges>: Exclude the specified ports from scanning
  -F: Fast mode - Scan fewer ports than the default scan
  -r: Scan ports consecutively - don't randomize
  --top-ports <number>: Scan <number> most common ports
  --port-ratio <ratio>: Scan ports more common than <ratio>
SERVICE/VERSION DETECTION:
  -sV: Probe open ports to determine service/version info
  --version-intensity <level>: Set from 0 (light) to 9 (try all probes)
  --version-light: Limit to most likely probes (intensity 2)
  --version-all: Try every single probe (intensity 9)
  --version-trace: Show detailed version scan activity (for debugging)
SCRIPT SCAN:
  -sC: equivalent to --script=default
  --script=<Lua scripts>: <Lua scripts> is a comma separated list of
        directories, script-files or script-categories
  --script-args=<n1=v1,[n2=v2,...]>: provide arguments to scripts
  --script-args-file=filename: provide NSE script args in a file
  --script-trace: Show all data sent and received
  --script-updatedb: Update the script database.
  --script-help=<Lua scripts>: Show help about scripts.
        <Lua scripts> is a comma-separated list of script-files or
        script-categories.
OS DETECTION:
  -O: Enable OS detection
  --osscan-limit: Limit OS detection to promising targets
  --osscan-guess: Guess OS more aggressively
TIMING AND PERFORMANCE:
  Options which take <time> are in seconds, or append 'ms' (milliseconds),
  's' (seconds), 'm' (minutes), or 'h' (hours) to the value (e.g. 30m).
  -T<0-5>: Set timing template (higher is faster)
  --min-hostgroup/max-hostgroup <size>: Parallel host scan group sizes
  --min-parallelism/max-parallelism <numprobes>: Probe parallelization
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  --min-rtt-timeout/max-rtt-timeout/initial-rtt-timeout <time>: Specifies
      probe round trip time.
  --max-retries <tries>: Caps number of port scan probe retransmissions.
  --host-timeout <time>: Give up on target after this long
  --scan-delay/--max-scan-delay <time>: Adjust delay between probes
  --min-rate <number>: Send packets no slower than <number> per second
  --max-rate <number>: Send packets no faster than <number> per second
FIREWALL/IDS EVASION AND SPOOFING:
  -f; --mtu <val>: fragment packets (optionally w/given MTU)
  -D <decoy1,decoy2[,ME],...>: Cloak a scan with decoys
  -S <IP_Address>: Spoof source address
  -e <iface>: Use specified interface
  -g/--source-port <portnum>: Use given port number
  --proxies <url1,[url2],...>: Relay connections through HTTP/SOCKS4 proxies
  --data <hex string>: Append a custom payload to sent packets
  --data-string <string>: Append a custom ASCII string to sent packets
  --data-length <num>: Append random data to sent packets
  --ip-options <options>: Send packets with specified ip options
  --ttl <val>: Set IP time-to-live field
  --spoof-mac <mac address/prefix/vendor name>: Spoof your MAC address
  --badsum: Send packets with a bogus TCP/UDP/SCTP checksum
OUTPUT:
  -oN/-oX/-oS/-oG <file>: Output scan in normal, XML, s|<rIpt kIddi3,
     and Grepable format, respectively, to the given filename.
  -oA <basename>: Output in the three major formats at once
  -v: Increase verbosity level (use -vv or more for greater effect)
  -d: Increase debugging level (use -dd or more for greater effect)
  --reason: Display the reason a port is in a particular state
  --open: Only show open (or possibly open) ports
  --packet-trace: Show all packets sent and received
  --iflist: Print host interfaces and routes (for debugging)
  --append-output: Append to rather than clobber specified output files
  --resume <filename>: Resume an aborted scan
  --stylesheet <path/URL>: XSL stylesheet to transform XML output to HTML
  --webxml: Reference stylesheet from Nmap.Org for more portable XML
  --no-stylesheet: Prevent associating of XSL stylesheet w/XML output
MISC:
  -6: Enable IPv6 scanning
  -A: Enable OS detection, version detection, script scanning, and traceroute
  --datadir <dirname>: Specify custom Nmap data file location
  --send-eth/--send-ip: Send using raw ethernet frames or IP packets
  --privileged: Assume that the user is fully privileged
  --unprivileged: Assume the user lacks raw socket privileges
  -V: Print version number
  -h: Print this help summary page.
EXAMPLES:
  nmap -v -A scanme.nmap.org
  nmap -v -sn 192.168.0.0/16 10.0.0.0/8
  nmap -v -iR 10000 -Pn -p 80
SEE THE MAN PAGE (https://nmap.org/book/man.html) FOR MORE OPTIONS AND EXAMPLES
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